EXP NO: 4 **CREATE UDF IN PIG**

$start-all.sh

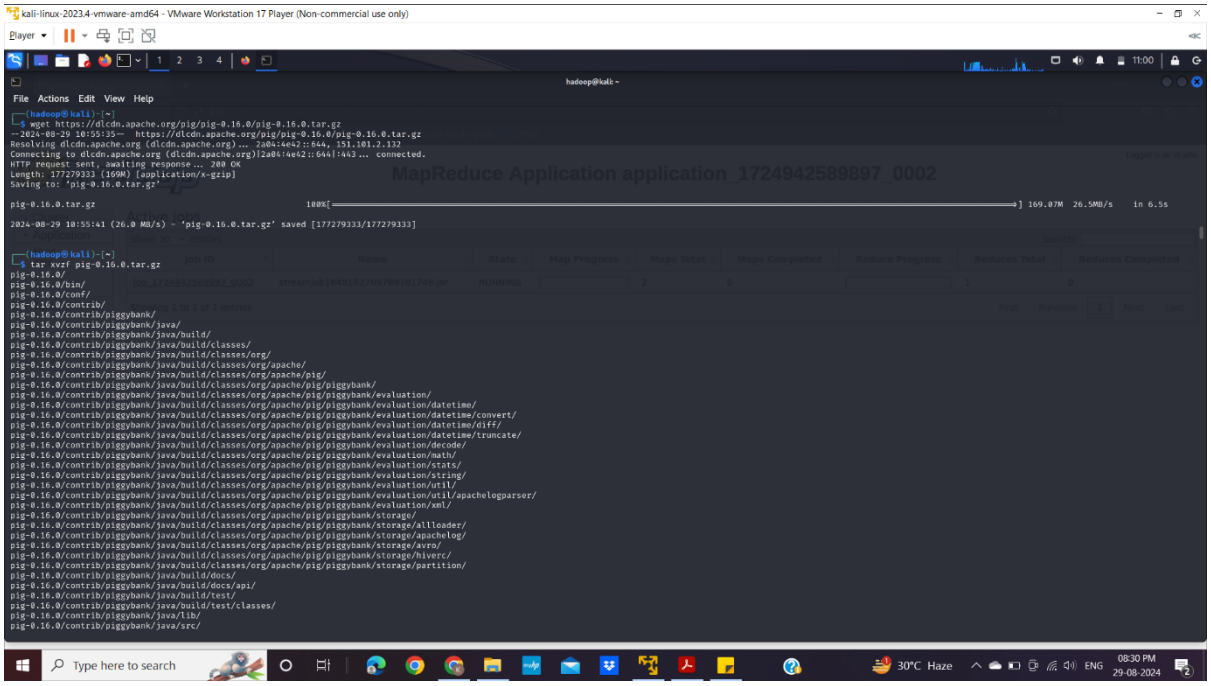
```
hadoop@kali: ~
File  Actions  Edit  View  Help

┌──(hadoop㉿kali)-[~]
└─$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [kali]
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
2024-09-11 04:59:16,429 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers
```

$ jps

```
┌──(hadoop㉿kali)-[~]
└─$ jps
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
14436 NodeManager
16772 Jps
13830 SecondaryNameNode
14311 ResourceManager
13597 DataNode
13471 NameNode
```

$wget https://dlcdn.apache.org/pig/pig-0.16.0/pig-0.16.0.tar.gz

$ tar xvzf pig-0.16.0.tar.gz

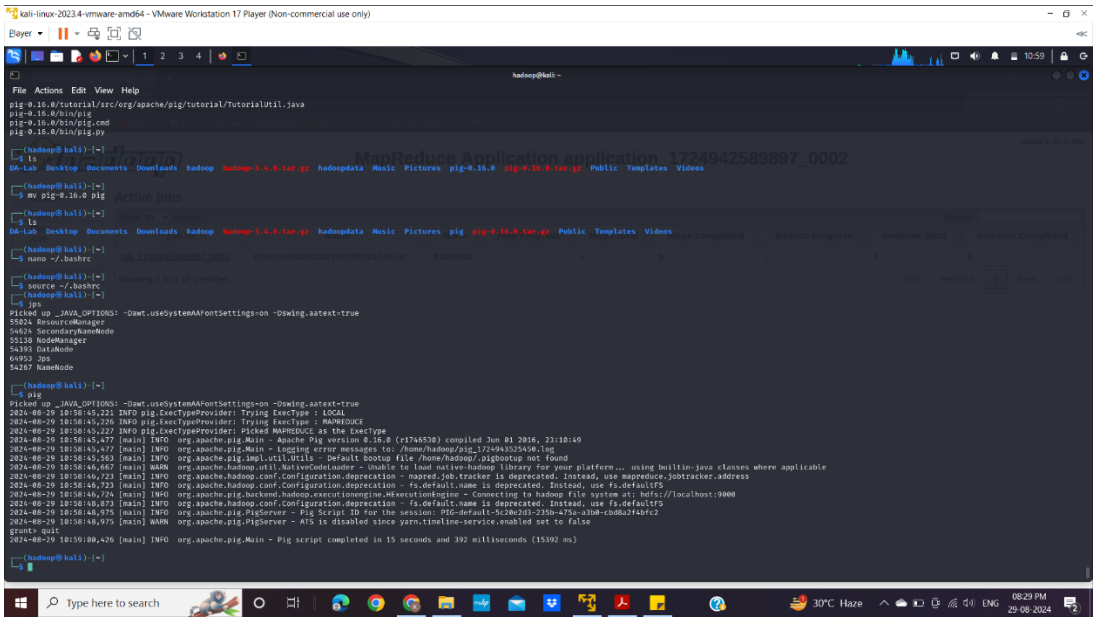


$nano ~/.bashrc

```
#PIG settings
export PIG_HOME=/home/hadoop/pig
export PATH=$PATH:$PIG_HOME/bin
export PIG_CLASSPATH=$PIG_HOME/conf:$HADOOP_HOME/etc/hadoop/
export PIG_CONF_DIR=$PIG_HOME/conf
#export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PIG_CLASSPATH=$PIG_CONF_DIR:$PATH
#PIG setting ends
```

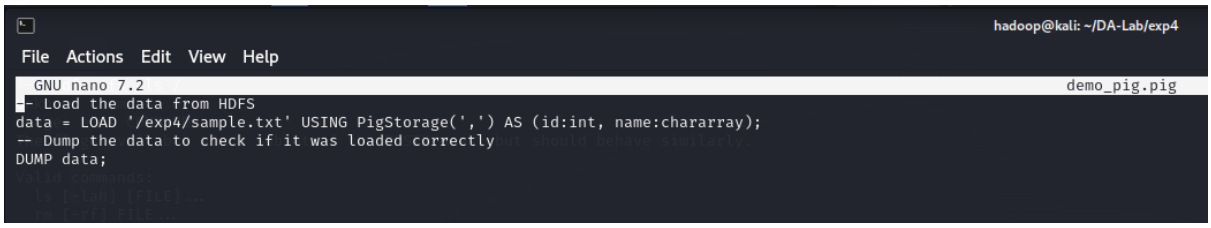
$mv pig-0.16.0 pig

$pig



$cd DA-Lab
$mkdir exp4
$cd exp4
$nano sample.txt

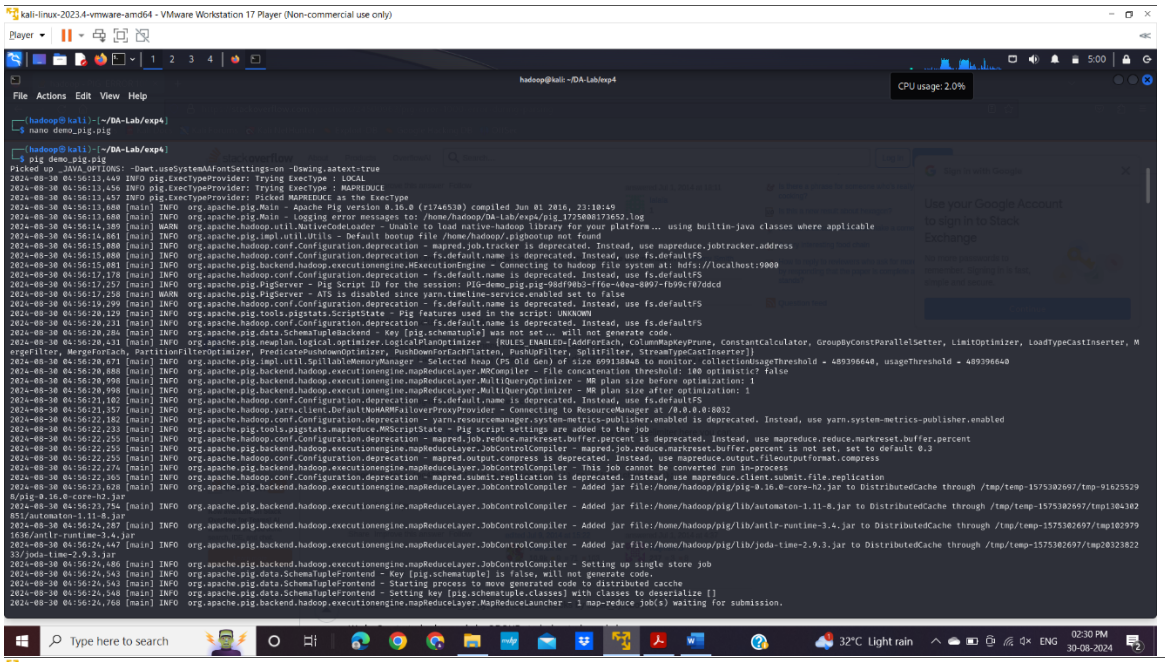
$nano demo_pig.pig

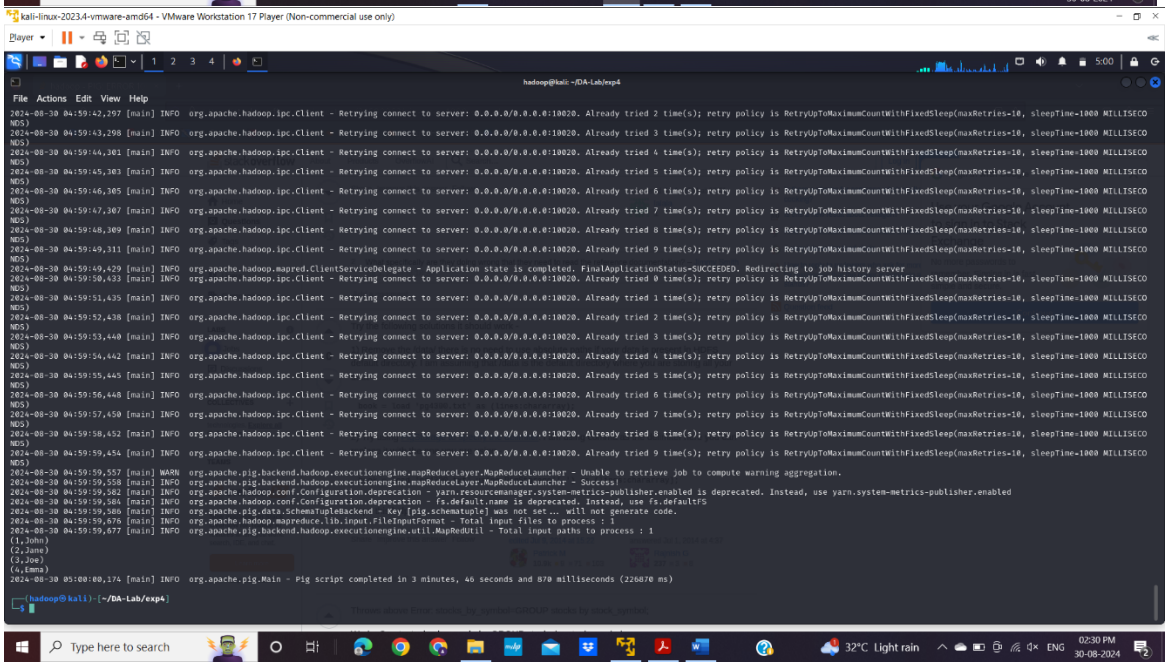


$hdfs dfs -mkdir /exp4

$hdfs dfs -copyFromLocal ~/DA-Lab/exp4/sample.txt /exp4

$pig demo_pig.pig





$nano uppercase_udf.py

$hdfs dfs -copyFromLocal ~/DA-Lab/exp4/uppercase_udf.py /exp4

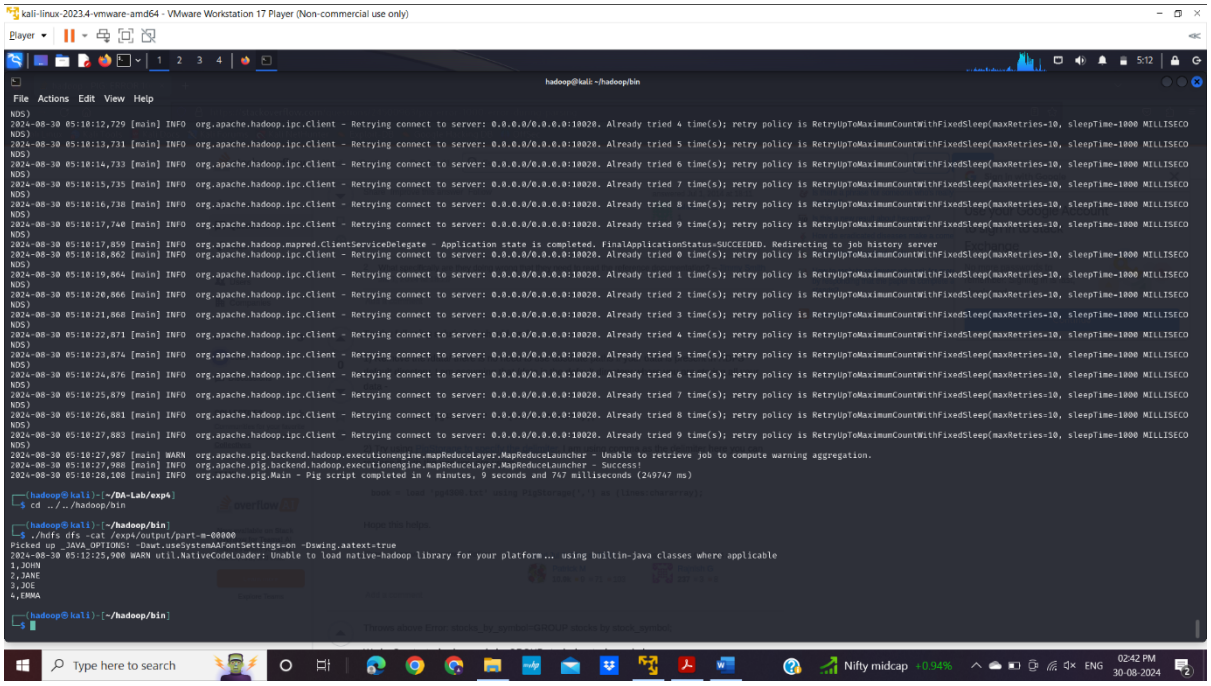
```
┌──(hadoop㉿kali)-[~/hadoop/bin]
└─$ ./hdfs dfs -ls /exp4
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
2024-09-21 00:26:01,736 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
Found 3 items
drwxr-xr-x   - hadoop supergroup          0 2024-08-30 05:07 /exp4/output
-rw-r--r--   1 hadoop supergroup         27 2024-08-30 04:43 /exp4/sample.txt
-rw-r--r--   1 hadoop supergroup        172 2024-08-30 05:02 /exp4/uppercase_udf.py
```

$nano udf_example.pig

```
hadoop@kali: ~/DA-Lab/exp4
File  Actions  Edit  View  Help
  GNU nano 7.2                                                    udf_example.pig
-- Register the Python UDF script
REGISTER 'hdfs:///exp4/uppercase_udf.py' USING jython AS udf;
-- Load some data
data = LOAD 'hdfs:///exp4/sample.txt' AS (text:chararray);
-- Use the Python UDF
uppercased_data = FOREACH data GENERATE udf.uppercase(text) AS uppercase_text;
-- Store the result
STORE uppercased_data INTO 'hdfs:///exp4/output';
```

$pig -f udf_example.pig

```
kali-linux-2023.4-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
hadoop@kali: ~/DA-Lab/exp4
File  Actions  Edit  View  Help
┌──(hadoop㉿kali)-[~/DA-Lab/exp4]
└─$ nano uppercase_udf.py

┌──(hadoop㉿kali)-[~/DA-Lab/exp4]
└─$ cd ../../hadoop/bin

┌──(hadoop㉿kali)-[~/hadoop/bin]
└─$ ./hdfs dfs -put ~/DA-Lab/exp4/uppercase_udf.py /exp4
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
2024-08-30 05:02:54,541 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

┌──(hadoop㉿kali)-[~/hadoop/bin]
└─$ cd ../../DA-Lab/exp4

┌──(hadoop㉿kali)-[~/DA-Lab/exp4]
└─$ nano udf_example.pig

┌──(hadoop㉿kali)-[~/DA-Lab/exp4]
└─$ pig -f udf_example.pig
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
2024-08-30 05:06:18,591 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2024-08-30 05:06:18,601 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2024-08-30 05:06:18,601 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2024-08-30 05:06:18,836 [main] INFO  org.apache.pig.Main - Apache Pig version 0.16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2024-08-30 05:06:18,836 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/hadoop/DA-Lab/exp4/pig_1725008778810.log
2024-08-30 05:06:19,685 [main] WARN  org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2024-08-30 05:06:20,127 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/hadoop/.pigbootup not found
2024-08-30 05:06:20,382 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2024-08-30 05:06:20,383 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:20,383 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2024-08-30 05:06:22,438 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:22,513 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-udf_example.pig-1dec0dcc-3e11-47e3-a9e8-561b5bb638a1
2024-08-30 05:06:22,513 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2024-08-30 05:06:22,725 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:24,290 [main] INFO  org.apache.pig.scripting.jython.JythonScriptEngine - created tmp python.cachedir=/tmp/pig_jython_3781842655894475147
2024-08-30 05:06:36,010 [main] WARN  org.apache.pig.scripting.jython.JythonScriptEngine - pig.cmd.args.remainders is empty. This is not expected unless on testing.
2024-08-30 05:06:36,044 [main] INFO  org.apache.pig.scripting.jython.JythonScriptEngine - Register scripting UDF: udf.uppercase
2024-08-30 05:06:37,784 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:37,867 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:38,407 [main] INFO  org.apache.pig.scripting.jython.JythonFunction - No schema defined for function 'uppercase' in /tmp/pig2315342629392933005tmp/uppercase_udf.py
2024-08-30 05:06:38,583 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:38,743 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2024-08-30 05:06:38,830 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2024-08-30 05:06:38,950 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-08-30 05:06:38,982 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2024-08-30 05:06:39,132 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, M
ergeFilter, MergeForEach, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2024-08-30 05:06:39,403 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699138048 to monitor. collectionUsageThreshold = 489396640, usageThreshold = 489396640
2024-08-30 05:06:39,684 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2024-08-30 05:06:39,858 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2024-08-30 05:06:39,859 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2024-08-30 05:06:39,994 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated. Instead, use fs.defaultFS
Type here to search
Live
ENG
02:37 PM
30-08-2024
```



$hdfs dfs -cat /exp4/output/*

```
┌──(hadoop㉿kali)-[~/hadoop/bin]
└─$ ./hdfs dfs -cat /exp4/output/*
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
2024-09-21 00:33:32,731 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
1,JOHN
2,JANE
3,JOE
4,EMMA
```